

STACK: Stack is a linear data structure that follows LIFO (Last In First Out).

PUSH: Insert element at TOP. POP: Remove element from TOP.

OVERFLOW: Stack is full. UNDERFLOW: Stack is empty.

QUEUE: Queue follows FIFO (First In First Out).

ENQUEUE: Insert at rear. DEQUEUE: Remove from front.

CIRCULAR QUEUE: Last position connects to first.

PRIORITY QUEUE: Elements served based on priority.

INFIX TO POSTFIX: Use stack, follow operator precedence.

EXAMPLE: $A+B*C \rightarrow ABC*+$

RECURSION: Function calling itself with base case.

TOWER OF HANOI: Moves = $2^n - 1$. For $n=3$, moves=7.

LOGIC: Move $n-1$ disks, move largest, move $n-1$ again.

TREE: Hierarchical data structure with root node.

LEAF NODE: Node with no children.

BINARY TREE: Max 2 children per node.

TRAVERSALS: Inorder (LNR), Preorder (NLR), Postorder (LRN).

BST: Left < Root < Right.

INSERT LOGIC: Compare and go left/right.

DELETE CASES: Leaf, one child, two children.

GRAPH: Set of vertices and edges.

BFS: Uses queue, level by level.

DFS: Uses stack/recursion, depth first.

PRIM'S: Choose smallest edge to form MST.

DIJKSTRA: Find shortest path using minimum distance.

LOGIC QUESTION: Ticket booking -> Queue (FIFO).

LOGIC QUESTION: Browser back -> Stack (LIFO).

LOGIC QUESTION: Expression evaluation -> Stack.

LOGIC QUESTION: Shortest path -> Dijkstra.

LOGIC QUESTION: Network connection -> Graph.